

Degeneracy, invariance, and exchange: proposed additions to v0.5

Editorial purpose and status. When does a geometrically degenerate triplet sector support closed, symmetry-compatible interaction dynamics? A threefold linear eigenvalue, an isotropic projected coupling, and a mode with low radiation are distinct properties. The existing counterexamples in v0.5 can make this distinction the central contribution. The two propositions below are finite-dimensional linear algebra, with no novelty claim. They introduce no new numerical results. The effective two-mode example in the final section motivates a future question; it is not a derived model of the project's Q-balls.

The source proposal is `BEITRAG.txt`, sections A–C, supplied by ag-phy-coordination on 30 September 2026. An independent algebraic reading within OpenAI is documented in `ALGEBRA-REVIEW.txt`; it is not an external scientific review. The original v0.5 manuscript and its artifacts remain unchanged.

1. Eliminated modes and the symmetry of a projected sector

Let $H = H^\dagger$ be a time-independent operator on a finite-dimensional complex Hilbert space. Let $P = P^\dagger = P^2$ be an orthogonal rank-three projector, and $Q = I - P$. The projectors select subspaces; P is not assumed to be a spectral projector of the full coupled H . In orthonormal bases of $\text{ran } P$ and $\text{ran } Q$, write

$$H = \begin{pmatrix} \lambda I_3 & V \\ V^\dagger & D \end{pmatrix}, \quad \lambda \in \mathbb{R}, \quad D = D^\dagger, \quad V = PHQ. \quad (\text{S1})$$

Here and below, block operators are restricted to their respective subspaces. Norms are Euclidean vector norms and their induced operator norms.

Proposition 1 (Schur correction and sufficient symmetry protection). *For real $z \notin \text{spec}(D)$, define*

$$H_{\text{eff}}(z) = \lambda I_3 - V(D - zI_Q)^{-1}V^\dagger. \quad (\text{S2})$$

Then

$$\det(H - zI) = \det(D - zI_Q) \det(H_{\text{eff}}(z) - zI_3). \quad (\text{S3})$$

Thus, away from the excluded poles, the full eigenvalue problem is equivalent to the reduced eigenvalue condition. Writing $\delta = \text{dist}(z, \text{spec}(D)) > 0$, one has

$$\|H_{\text{eff}}(z) - \lambda I_3\| \leq \frac{\|V\|^2}{\delta}. \quad (\text{S4})$$

If a unitary group representation $R(g)$ satisfies $[R(g), H] = [R(g), P] = 0$ for every g , and its restriction to $\text{ran } P$ is irreducible over $\mathbb{C}$, then $H_{\text{eff}}(z)$ is scalar on $\text{ran } P$.

Proof. For a full eigenvector (u, v) , the lower block equation gives $v = -(D - zI_Q)^{-1}V^\dagger u$. Substitution into the upper block yields $(H_{\text{eff}}(z) - zI_3)u = 0$. At the allowed values of z , a nonzero full eigenvector cannot have $u = 0$, and each reduced eigenvector reconstructs a full one by this formula. Block elimination gives (S3). Since D is Hermitian, the spectral theorem gives $\|(D - zI_Q)^{-1}\| = 1/\delta$, proving (S4).

Commutation with P makes $R(g)$ block diagonal, with restrictions $R_P(g)$ and $R_Q(g)$. Commutation with H implies $R_P V = V R_Q$ and $[R_Q, D] = 0$. Consequently $V(D - zI_Q)^{-1}V^\dagger$ commutes with every $R_P(g)$. Schur's lemma on the irreducible complex triplet then makes this operator, and hence $H_{\text{eff}}(z)$, scalar. $\square$

Explicit counterexample to protection by projection alone. Take a single complementary mode $D = \Delta$, with $\Delta \in \mathbb{R}$, and $V = g(1, 0, 0)^T$, $g \in \mathbb{C} \setminus \{0\}$. For $z \neq \Delta$,

$$H_{\text{eff}}(z) - \lambda I_3 = -\frac{|g|^2}{\Delta - z} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{S5})$$

One triplet direction is distinguished even though the direct compression PHP is exactly isotropic. For real g , the numerator is g^2 . Equal eigenvalues of PHP therefore do not by themselves protect the symmetry of the reduced coupled problem.

Limits of the statement. The determinant reduction is not to be evaluated at a pole $z \in \text{spec}(D)$. The norm estimate controls the entire correction, not a compulsory amount of anisotropy; a nonzero correction can be isotropic. A full symmetry representation can protect isotropy while still allowing transfer to equivalent representations inside Q . It does not in itself imply $QHP = 0$ or absence of leakage. Moreover, $H_{\text{eff}}(z)$ is an energy-dependent Schur operator, not automatically a time-local generator for the projected evolution. The actual Q-ball frequency pencil and its normalization must be derived and treated on their own terms. This finite Schrödinger-type formula cannot be imported into that problem without justification.

2. A conservative finite-time leakage bound

Proposition 2 (Leakage from an initially occupied subspace). *For the same time-independent Hermitian H , let $i\dot{\psi} = H\psi$, $\|\psi(0)\| = 1$, and $Q\psi(0) = 0$. Then, for every real t ,*

$$Q\psi(t) = -i \int_0^t e^{-iD(t-s)} QHP P\psi(s) ds, \quad (\text{S6})$$

and

$$\|Q\psi(t)\| \leq \min\{1, |t| \|QHP\|\}, \quad \|Q\psi(t)\|^2 \leq \min\{1, t^2 \|QHP\|^2\}. \quad (\text{S7})$$

These statements hold for any orthogonal projector P , not only rank three.

Proof. Projecting the evolution gives $i d(Q\psi)/dt = DQ\psi + QHP P\psi$. Variation of constants, with zero initial Q -component, yields (S6). Both the full evolution and $e^{-iD(t-s)}$ are unitary. Thus $\|P\psi(s)\| \leq 1$, and the integral has norm at most $|t| \|QHP\|$. For negative t , reverse the oriented integration interval; its length is still $|t|$. The full-state normalization independently gives $\|Q\psi(t)\| \leq 1$. Combining the two bounds and squaring proves (S7). $\square$

Connection to the existing invariance checks. For classical complex amplitudes, $\|Q\psi(t)\|^2$ is the fraction of the normalized squared norm outside the chosen subspace. Calling it a probability additionally presumes a quantum-state interpretation. The exact condition $QHP = 0$ implies zero leakage for every initial state in P . For an observation window $|t| \leq T$, $T > 0$, a sufficient condition for an outside norm fraction at most $\varepsilon \in [0, 1]$ is $\|QHP\| \leq \sqrt{\varepsilon}/T$. Reported Frobenius norms of outside blocks may be used as conservative upper bounds on the induced norm, provided the projector, normalization, and block definition actually match. This connects the algebraic invariance criterion to a finite tolerated error. It proves neither sharp long-time stability nor an optimal lifetime. For large times the bound may reduce to the trivial value one. Nonlinear fields require other error estimates; no nonlinear or continuum certificate follows from (S7).

3. A future question: dark and still interacting?

As a purely effective passive two-mode illustration, consider

$$\mathcal{H} = \begin{pmatrix} \omega - i\gamma/2 & J - i\gamma_{12}/2 \\ J - i\gamma_{12}/2 & \omega - i\gamma/2 \end{pmatrix}, \quad \omega, J, \gamma, \gamma_{12} \in \mathbb{R}, \quad \gamma \geq |\gamma_{12}|. \quad (\text{S8})$$

The last condition makes the loss matrix positive semidefinite. The normalized symmetric and antisymmetric vectors have eigenvalues

$$z_{\pm} = \omega \pm J - \frac{i}{2}(\gamma \pm \gamma_{12}). \quad (\text{S9})$$

For $\gamma_{12} = \gamma$, the antisymmetric channel has zero decay in this model even when $J \neq 0$. Thus zero decay in a channel is not logically equivalent to absent coherent coupling. This is a familiar algebraic possibility, not a parameter choice derived from KRAFT or from the project’s microscopic field model.

In a microscopic mediation mechanism, the dispersive and dissipative parts arise from the same causal response and cannot simply be chosen independently. Nor does a fixed J establish a spatial force: that requires positional dependence, dynamical positions, and consistent energy and momentum accounting. The coordinator reports that KRAFT-still failed its strict force-switch criterion; this example does not reinterpret that result.

The smallest proposed next question is whether the same field or scattering operator yields both finite coherent frequency splitting and small radiation, with a consistently defined distance force where applicable. Frequency splitting, radiation channels, and energy flux must refer to the same configuration. Different phase windows or fits must not be presented as one joint finding. This is a proposed study, not a calculation performed for this supplement.

4. Placement and boundaries

Propositions 1 and 2 are proposed for insertion after the exchange and subspace-invariance criteria of v0.5, section 4. They formalize why a degenerate compression or an isotropic projected matrix is insufficient when complementary modes remain coupled. Section 3 belongs among the concrete future field-model questions, not among the already completed triplet experiments.

Frequency-dependent mediator range, Hamiltonian phase memory, and Hawking-related questions are retained in the separate, attributed `RESEARCH-NOTE-EN.txt`. They do not extend the claims of the main paper. No KRAFT numbers are mixed into the triplet results, and the existing BIC/exterior proof remains a separate line of work.